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Roll No.:

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Midterm (Missed) Examination – May 2024

Second Semester

Integrated MSc Physics with Minor in Scientific Computing/Data Science

22PHY114 Electricity and Magnetism

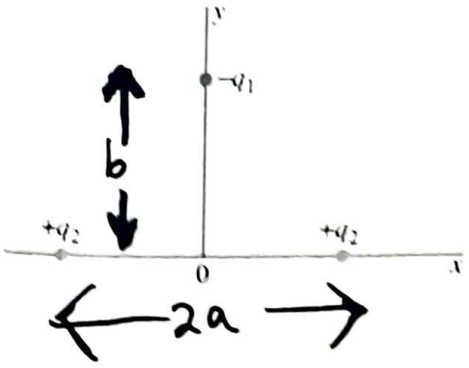
Graph not required

Time :2 hours

Maximum Marks:50

Instruction

1. Answer all the questions.
2. Answer with appropriate significant figures and units.

Qn No.		Marks	CO	Blooms Taxonomy level
1	Two capacitors when connected in series have an equivalent capacitance of 5.5 μF. If the two capacitors are connected in series, the equivalent capacitor is 35 μF. Find the capacitance of each capacitor.	7	CO2	L3
2	Two positive charges (indicated as $+q_2$) are placed apart by a distance $2a$. In the middle of the charges is the origin. On the positive y-axis, there is a negative charge (indicated by $-q_1$) that is initially at rest at a distance b from the origin. What is the minimum velocity that needs to be given to the negative charge so that it escapes to an infinitely large distance from the two positive charges? 	7	CO2	L3

3	A solid metal cube has a spherical cavity at its centre. At the centre of the cavity there is a point charge Q = +14.0 μC. The metal cube carries a net charge −5.0 μC (not including Q). a) Determine the total charge on the surface of the spherical cavity. b) Determine the total charge on the outer surface of the cube.	7	CO1	L3
4	There are two parallel plate conductors such that the potential as function of position, x is given by $V(x) = \alpha x + 5$ with $x = 0$ taken at one of the plates and x positive in the direction of the other plate. α is a positive constant. All the quantities are in SI units. a) What is the electric field in between the plates? b) What is the charge density on the plates?	7	CO2	L3
5	Two negative charges ($-Q_0$ and $-4Q_0$) are separated by a distance L. There is a third charge placed in such a way so that the first two charges are not moving and in equilibrium. a) Find the location where the third charge needs to be placed. b) What is the sign of the third charge? c) What is the magnitude of the third charge?	8		
6	An electron with initial velocity 8.0×10^4 m/s along positive y-axis enters a region of electric field **E** where $\mathbf{E} = (2.0\,\mathbf{i} + 8.0\,\mathbf{j}) \times 10^4$ N/C. a) Find the acceleration of the electron as a function of time. b) At 1.0×10^{-9} s, what is the angle that the electron moves relative to its initial direction?	7	CO1	L3
7	A charge Q is uniformly distributed on a thin wire of length L. This wire is in the shape of semicircle. Find the electric potential at the centre of the semicircle. You can assume electric potential is zero at large distances from the wire.	7	CO2	L3